

Enhancing Judgment Document Generation via Agentic Legal Information Collection and Rubric-Guided Optimization

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Abstract

Automating the drafting of judgment documents is pivotal to judicial efficiency, yet it remains challenging due to the dual requirements of comprehensive retrieval of legal information and rigorous logical reasoning. Existing approaches, typically relying on standard Retrieval-Augmented Generation and Supervised Fine-Tuning, often exhibit insufficient evidence recall and hallucinated statutory references. To bridge this gap, we propose Judge-R1, a unified framework that enhances generation through two key mechanisms. First, we introduce Agentic Legal Information Collection, which employs a dynamic planning agent to retrieve precise statutes and precedents from multiple sources. Second, we implement Rubric-Guided Optimization, a reinforcement learning phase utilizing Group Relative Policy Optimization (GRPO) with a comprehensive legal reward function to enforce adherence to judicial standards and reasoning logic. Extensive experiments on the JuDGE benchmark demonstrate that Judge-R1 significantly outperforms state-of-the-art baselines in both legal accuracy and generation quality¹.

1 Introduction

In recent years, the rapid evolution of Large Language Models (LLMs) has catalyzed transformative applications within the legal domain, ranging from contract analysis and case law research to automated document drafting [1, 14, 19]. Among these, judgment document generation stands out as a critical yet exceptionally challenging Information Retrieval (IR) and Generation task. Judgment documents are authoritative texts that encapsulate complex legal reasoning, procedural nuances, and final rulings, serving as the cornerstone of judicial proceedings [9]. In practice, drafting these documents is labor-intensive, requiring judges to synthesize vast amounts of legal information, including statutes, precedents, and jurisprudence, into a coherent, legally sound judgment document. Consequently, automating this process holds immense potential to alleviate judicial burdens and enhance procedural efficiency.

However, despite the promise of generative AI, high-quality judgment document generation remains an open challenge due to two fundamental bottlenecks: comprehensive information collection

and legally grounded generation. First, the Information Collection stage acts as the foundation of legal reasoning. To draft a valid judgment, a system must accurately retrieve the specific statutes and relevant case precedents that form the "Major Premise" of the legal syllogism. This is non-trivial because legal queries are often verbose and abstract (derived from raw case facts), and the retrieval space is riddled with noise (e.g., distinguishing between visually similar but legally distinct statutes). Failure to recall a key statute or relevant precedent inevitably results in a legally flawed output, regardless of the model's capabilities. Second, the Document Generation stage demands more than linguistic fluency. It requires the systematic integration of the retrieved legal knowledge with the factual "Minor Premise" to deduce a correct conclusion. The generated text must strictly adhere to judicial standards regarding sentencing logic, structural integrity, and professional tone.

Existing approaches often fall short in addressing these dual challenges. Regarding information collection, standard Retrieval-Augmented Generation (RAG) methods typically rely on semantic matching, which lacks the planning capability to decompose complex case facts into precise legal queries, often resulting in insufficient recall or irrelevant noise. For the generation process, the prevailing paradigm relies heavily on Supervised Fine-Tuning (SFT) [18]. While SFT enables models to mimic the stylistic surface of legal texts, it fundamentally relies on imitation learning. It does not explicitly penalize logical fallacies or incentivize the model to self-correct towards a better judgment. Consequently, models often hallucinate citations or produce logically incoherent sentences that look professional but are legally unsound.

To this end, we introduce Judge-R1, a novel framework that bridges the gap between complex legal retrieval and rigorous judgment generation. Judge-R1 addresses these limitations through a synergistic approach. For Information Collection, we propose an Agentic Legal Information Collection mechanism. Unlike static retrievers, our method employs a Multi-source Retrieval-Augmented Generation (MRAG) system driven by an intelligent agent. This agent dynamically plans search queries to cover diverse legal points (e.g., constitutive elements, sentencing factors) and employs a multi-stage filter to construct a high-quality evidence set comprising both statutes and precedents. For Judgment Document Generation, we move beyond simple imitation by introducing a Reinforcement

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¹All the codes and datasets are available at: github.com/oneal2000/Judge-R1

Learning (RL) alignment phase specifically designed for the legal domain. After a warm-up stage using SFT, we implement Group Relative Policy Optimization (GRPO) to further refine the model [5]. Crucially, we design a comprehensive legal rubric as the reward function, which evaluates the output across multiple dimensions: legal accuracy (e.g., correct statutes and sentencing), textual quality, and reasoning depth. This allows the model to self-optimize its generation policy, learning to produce judgments that are not only linguistically coherent but also legally robust.

Extensive experiments on the JuDGE benchmark demonstrate that Judge-R1 outperforms strong baselines. For Information Collection, our agentic framework provides superior retrieval performance compared with standard RAG systems. For Judgment Document Generation, our RL-aligned approach yields judgments with significantly higher legal accuracy and logical consistency than existing methods. In summary, our contributions are:

- We propose Judge-R1, a unified framework that integrates agentic information collection with rubric-driven reinforcement learning for judgment document generation.
- Empirical results show that Judge-R1 achieves state-of-the-art performance, offering a principled path for improving the reliability of legal text generation.

2 Problem Formulation

We formalize judgment document generation as a two-stage task. Given a fact description f and a legal corpus $\mathcal{D} = \{d_1, d_2, \dots, d_N\}$ containing statutes and precedents, the goal is to generate a judgment document j that is legally sound and logically coherent. The process consists of:

- **Legal Information Collection:** A retrieval system identifies a relevant evidence set $\mathcal{E}_f \subseteq \mathcal{D}$ containing the specific statutes and precedents pertinent to the facts f .
- **Judgment Generation:** A generative system synthesizes the judgment j conditioned on the fact description f and the retrieved evidence $\mathcal{E}_f$, ensuring the output strictly adheres to judicial reasoning and sentencing standards.

3 Methodology

We propose **Judge-R1**, a framework designed to synthesize legally sound judgment documents j grounded in retrieved evidence $\mathcal{E}_f$. To ensure the generated judgments are both accurate in citing statutes and compliant with judicial reasoning, Judge-R1 operates through a three-stage pipeline (shown in Figure 1):

- (1) **Hybrid Information Collection:** A dual-route retrieval module that combines a standard dense backbone with an LLM-driven agent to construct a high-precision evidence set $\mathcal{E}_f$ from the corpus $\mathcal{D}$.
- (2) **Supervised Fine-Tuning (SFT):** The generator G_θ takes the instruction, facts f , and retrieved laws $\mathcal{E}_f$ as input to generate the full judgment document j , establishing the foundational structure and formatting conventions.
- (3) **Group Relative Policy Optimization (GRPO):** We further align the model using reinforcement learning with a multi-dimensional legal rubric, rewarding correctness and logic over mere surface-level mimicry.

3.1 Information Collection

Overview. We aim to retrieve a concise, high-recall set of applicable statutes $\mathcal{E}$ from fact description f . To balance coverage with precision, we employ a hybrid system fusing a robust neural backbone with a GRPO-enhanced reasoning agent.

Neural Backbone with Hard-Negative Mining. The foundation is a two-stage retrieval pipeline. A bi-encoder encodes q and d into embeddings for dot-product retrieval. Crucially, we enhance discrimination using contrastive learning with static hard negatives mined from top-ranked near neighbors (excluding positives). Top candidates are refined by a BERT-based cross-encoder. To align training with inference difficulty and prevent leakage, we construct training triples using a K-fold mining strategy, where negatives for one fold are retrieved by a model trained on disjoint folds.

GRPO-Enhanced Retrieval Agent. Standard retrieval struggles with verbose, multi-issue narratives. We introduce an agentic workflow optimized via Group Relative Policy Optimization (GRPO) [5] to align planning with downstream utility:

- (1) **QueryGen (Planning):** The agent decomposes f into m sub-queries targeting heterogeneous legal aspects (e.g., elements, sentencing). We apply GRPO here, rewarding the model for retrieval effectiveness (i.e., maximizing MRR and Recall@50 for the generated queries) rather than merely linguistic quality.
- (2) **Multi-View Recall & Rerank:** We aggregate top- k results from the backbone for all sub-queries to ensure high recall.
- (3) **LLM-Select (Filtering):** The agent acts as a final filter, outputting a selected set with inclusion/exclusion rationales. To prevent hallucination or over-pruning, we train this stage with GRPO using rewards that maximize early precision ($P@5$) and recall while penalizing set sizes outside a distinct range (5–10). A fallback mechanism ($N_{\min}=5$) supplements candidates if confidence is low.

Multi-route Fusion. Finally, we fuse the agentic results with a standard reranking approach using Reciprocal Rank Fusion (RRF). We assign higher weight to the agent ($w_{\text{agent}}=2.0$, $w_{\text{std}}=1.0$) to prioritize reasoned selection while maintaining robustness.

3.2 Judgment Document Generation

Overview. The generation module synthesizes a comprehensive judgment document j conditioned on the fact description f and the retrieved evidence set $\mathcal{E}_f$. The objective is to produce a document that not only reaches the correct verdict but also strictly adheres to judicial reasoning standards. To achieve this, we employ a two-stage optimization strategy: Supervised Fine-Tuning (SFT) for structural format alignment, followed by Group Relative Policy Optimization (GRPO) to enforce legal correctness and reasoning fidelity.

Supervised Fine-Tuning (SFT). We first adapt the base Large Language Model (LLM) to the judicial domain using retrieval-augmented instances. The input sequence is standardized as a concatenation of the instruction, retrieved laws $\mathcal{E}_f$, and facts f , while the target output is the complete judgment document j . We optimize the model using the standard next-token prediction loss, computing gradients on j . This stage primarily focuses on stabilizing the generation format, ensuring the model learns to organize the judgment j into

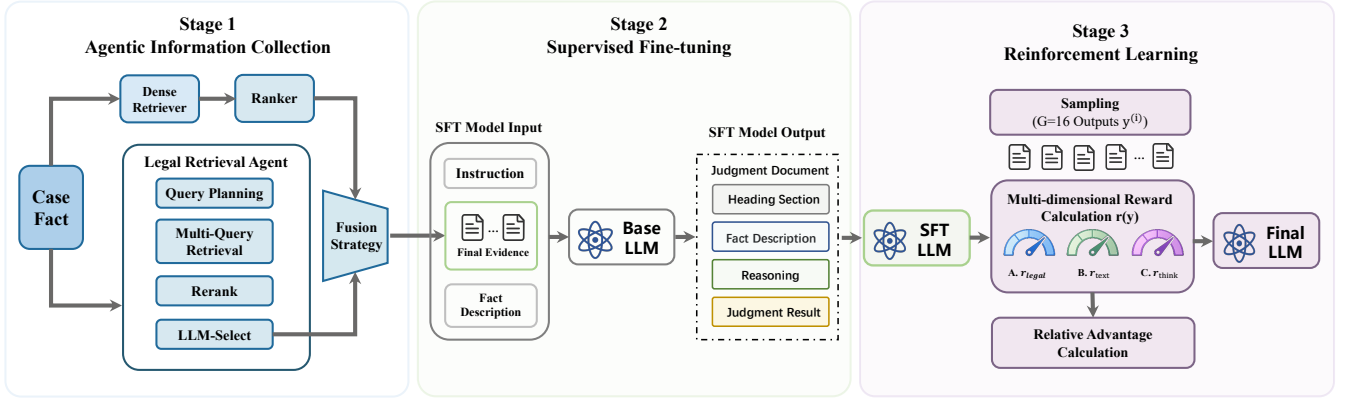


Figure 1: An illustration of our proposed framework.

distinct sections (e.g., fact acknowledgment, reasoning, and sentencing) and to explicitly cite the provided statutes. We use LoRA for parameter-efficient tuning, enabling the model to internalize the specific writing style of legal judgments while preserving its general reasoning capabilities. This warm-up phase provides a robust policy initialization for subsequent reinforcement learning.

Group Relative Policy Optimization (GRPO). While SFT establishes the document structure, it does not inherently guarantee the accuracy of the legal conclusion. To align the model with high-stakes judicial criteria, we further optimize the generator using GRPO. Unlike traditional PPO, which requires a separate value network, GRPO estimates the baseline from the group scores of sampled outputs, making it highly efficient for long-context generation. For each input, we sample a group of G candidate judgments $\{j^{(i)}\}_{i=1}^G$ from the current policy π_θ . The policy is updated to maximize the advantage of high-scoring candidates while maintaining stability via a KL-divergence constraint relative to the SFT reference model. The objective is formalized as maximizing the expected advantage $A^{(i)}$ derived from a comprehensive reward rubric:

$$A^{(i)} = \frac{r(j^{(i)}) - \mu_G}{\sigma_G + \epsilon}, \quad \text{where } \mu_G = \frac{1}{G} \sum_{k=1}^G r(j^{(k)}). \quad (1)$$

Rubric-Aware Reward Design. Standard reward models typically prioritize generic fluency, which is insufficient for the strict requirements of judicial adjudication. To bridge this gap, we collaborated with legal professionals to codify the criteria for high-quality judgment documents into a multidimensional rubric for evaluating quality. The total reward $r(j)$ is a weighted sum of three pillars:

$$r(j) = w_1 \cdot r_{\text{legal}}(j) + w_2 \cdot r_{\text{struct}}(j) + w_3 \cdot r_{\text{logic}}(j), \quad (2)$$

where the weights $w_{1,2,3}$ and internal sub-weights are configurable hyperparameters (see Sec. 4).

- **Legal Correctness (r_{legal}):** This component ensures the judgment is legally binding by evaluating extracted legal entities (statutes, crimes, sentencing terms, and fines) against the ground truth. Specifically, we calculate the $F1$ -score for the sets of statutory citations and charges. Following the setting of JuDGE [18],

Table 1: Retrieval performance for information collection. Metrics include Precision (P), and Recall (R) at various cutoffs (5 and 10). The best results are in bold.

	P@5	P@10	R@5	R@10
TF-IDF	0.1625	0.1200	0.2096	0.3074
BM25	0.1501	0.1024	0.1958	0.2649
Law Retriever	0.4535	0.3509	0.5529	0.8262
Agentic (Ours)	0.6383	0.3878	0.7806	0.9159

for numerical sentencing and fines, we employ a matching score:

$$S(A, B) = \max(0, 1 - \frac{|A - B|}{\max(A, B, 1)})$$

to penalize deviations, while assigning a zero score for categorical conflicts (e.g., acquittals vs. convictions).

- **Structural Professionalism (r_{struct}):** A legally sound verdict must adhere to canonical court formats. We segment the document into *reasoning* and *judgment* sections and compute the average BERTScore between the generated segments and their corresponding ground-truth references. This encourages both linguistic professionalism and structural integrity. Missing or unextractable section is assigned a score of 0 for that section.
- **Logical Consistency (r_{logic}):** To ensure the verdict stems from rigorous reasoning rather than "result-oriented" hallucination, we evaluate the Chain-of-Thought (CoT) within the `<think>` tags. Specifically, we define $r_{\text{logic}} = \min(1, S_{\text{len}} + S_{\text{rep}})$, where S_{len} and S_{rep} are piecewise step functions that reward moderate reasoning length and penalize structural redundancy (e.g., excessive bullet points), respectively. All interval thresholds and reward increments are treated as configurable hyperparameters (see Sec. 4) to discourage degenerate behaviors like circular reasoning or "thinking-shortcut" hallucinations.

4 Experimental Setup

Benchmark We conduct our experiments on the **JuDGE** benchmark [18], a dataset specifically constructed to evaluate the generation of complete judgment documents based on factual descriptions.

Table 2: Main experimental results. We report accuracy for Penalty (Prison and Fine terms), and Recall, Precision, and F1 scores for Convicting and Referencing tasks. For the Reasoning and Judgment sections, we use METEOR (MET.) and BERTScore (BERTS.) as metrics. The best results for each model are highlighted in bold.

Model	Method	Penalty Acc.		Convicting Acc.			Referencing Acc.			Reasoning Section		Judgment Section	
		Prison	Fine	Recall	Prec.	F1	Recall	Prec.	F1	MET.	BERTS.	MET.	BERTS.
QW-2.5-3B	Direct	0.6088	0.4639	0.8463	0.8426	0.8445	0.4666	0.7288	0.5689	0.4060	0.8059	0.3034	0.8030
	ICL	0.6332	0.4659	0.8792	0.8699	0.8730	0.4813	0.6785	0.5631	0.3890	0.7900	0.3895	0.8274
	SFT	0.6440	0.4971	0.8962	0.8985	0.8974	0.6130	0.7287	0.6659	0.5087	0.8696	0.5843	0.8872
	RAG+SFT	0.5825	0.4977	0.9102	0.9102	0.9102	0.5563	0.6167	0.5850	0.3896	0.7691	0.3776	0.8134
	Judge-R1	0.6625	0.4857	0.9281	0.9278	0.9280	0.7589	0.6905	0.7231	0.5660	0.8853	0.6558	0.9036
QW-3-4B	Direct	0.5409	0.3454	0.9142	0.9125	0.9133	0.4978	0.6668	0.5700	0.4508	0.8298	0.3976	0.8215
	ICL	0.5909	0.3973	0.9321	0.9391	0.9256	0.4981	0.7199	0.5888	0.3873	0.7622	0.4350	0.8627
	SFT	0.6224	0.4443	0.9082	0.9079	0.9080	0.5753	0.7216	0.6402	0.5051	0.8741	0.5455	0.8829
	RAG+SFT	0.5732	0.4648	0.9321	0.9358	0.9340	0.7062	0.6470	0.6753	0.4460	0.8098	0.5368	0.8719
	Judge-R1	0.6573	0.5144	0.9391	0.9408	0.9400	0.7739	0.7052	0.7379	0.5363	0.8855	0.6476	0.9042

The dataset comprises 2,505 fact-judgment pairs, officially partitioned into 2,004 training instances and 501 test instances.

Baselines For the baselines of the Information Collection stage, we employ traditional sparse retrievers including TF-IDF and BM25. We also include the Law Retriever [18], the previous state-of-the-art dense bi-encoder, which is explicitly trained via contrastive learning to align case facts with statutes. For the Judgment Generation stage, we adopt three baselines from the JuDGE benchmark [18]: (1) Few-shot ICL, which prompts LLMs with fact-judgment exemplars; (2) SFT, which directly optimizes the model on ground-truth documents; and (3) RAG+SFT, specifically the Multi-Source RAG (MRAG) framework that augments SFT with external knowledge from both Law and Case Retrievers.

Implementation Details We utilize Qwen2.5-3B-Instruct and Qwen3-4B-Thinking as our backbone models. The dense retriever is initialized with Chinese-RoBERTa-WWM and optimized using AdamW (LR=5e-6, fp16), trained on the training set from the JuDGE benchmark, while LLM fine-tuning employs a learning rate of 3e-5 (bf16). For the GRPO stage, we train for 1 epoch (LR=5e-6, KL $\beta = 0.05$) with 16 generations per input, using a reward function that aggregates Legal, Text, and Thinking scores with weights of 0.6, 0.3, and 0.1. All experiments are conducted on 8 NVIDIA A100 (40GB) GPUs using default Hugging Face configurations. Due to space constraints, comprehensive implementation details and source code are provided in our anonymous GitHub repository.

5 Experimental Results

5.1 Analysis of Legal Information Collection

The evaluation results on the Information Collection stage demonstrate the decisive advantage of our proposed Judge-R1 framework over traditional retrieval baselines. As indicated by the performance comparison, the Agentic Legal Information Collection mechanism substantially improves the retrieval of relevant legal evidence, including statutes and precedents. Two key trends emerge from the analysis. First, Judge-R1 exhibits significantly higher recall compared to both lexical matching methods and the previous SOTA dense retrieval method. This indicates that the agent’s ability to

dynamically decompose complex case facts into precise queries effectively captures obscure legal points that static semantic matching often overlooks. Second, the multi-stage filtering mechanism ensures high precision by effectively suppressing noise in the retrieval space and distinguishing between legally distinct but textually similar statutes. Overall, these observations validate that moving beyond passive retrieval to an agentic, planning-based approach is essential for handling complex legal reasoning tasks.

5.2 Analysis of Judgment Document Generation

As shown in Table 2, Judge-R1 consistently outperforms all baselines across both model backbones, validating the effectiveness of our synergistic framework. First, regarding legal decision-making (Penalty and Convicting), Judge-R1 achieves substantial gains over the strong SFT baseline. This indicates that while SFT captures surface-level stylistic patterns, our RL alignment via GRPO effectively reinforces logical rigor, enabling the model to deduce more accurate sentencing terms and conviction results. Second, with respect to citation and evidence utilization (Referencing), our method outperforms standard RAG approaches. The superior Precision and F1 scores demonstrate that the proposed Agentic Legal Information Collection mechanism reduces retrieval noise and aligns retrieved evidence more precisely with case facts than varying semantic matching. Finally, for document generation quality (Reasoning and Judgment sections), Judge-R1 secures the highest scores in both METEOR and BERTScore. This confirms that our framework not only improves legal accuracy but also enhances the structural integrity and coherence of the generated text, successfully synthesizing complex legal logic into professional-grade judgments.

6 Related Work

Early research in legal intelligence primarily focused on Legal Judgment Prediction (LJP), which treats judicial reasoning as a classification task to predict charges, articles, or sentencing terms based on case facts [2, 6, 7, 10–12]. While effective for prediction, these methods cannot automate the drafting of complete, authoritative judicial documents. To address this, the JuDGE benchmark [18]

recently introduced the task of Judgment Document Generation, employing Large Language Models (LLMs) for automatic judgment document generation. However, existing baselines largely depend on standard Retrieval-Augmented Generation (RAG) [3, 16, 17] and Supervised Fine-Tuning (SFT). These approaches often struggle to retrieve precise statutes [4, 8, 13, 15] due to semantic ambiguity and lack the objective-driven feedback needed to prevent hallucinations in complex legal reasoning. To bridge these gaps, our Judge-R1 framework integrates agentic planning for precise information collection and rubric-driven reinforcement learning to ensure legal rigor and consistency.

7 Conclusion

In this paper, we presented **Judge-R1**, a novel framework that integrates agentic legal information collection with rubric-driven reinforcement learning to improve judgment document generation. Our method addresses bottlenecks in judgment document generation by enhancing the fidelity of legal information collection and generation, achieving state-of-the-art performance on the JuDGE benchmark. In the future, we aim to generalize this paradigm to broader legal reasoning tasks and explore more efficient alignment strategies for the legal domain.

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